Driver–Circuit Compatibility in Formula 1: A Hybrid Unsupervised–Supervised Approach to Predicting Race Performance

Yash Lalwani
Department of Computer Science
yl2661@scarletmail.rutgers.edu

Hanumath Mandadi
Department of Computer Science
hcm38@scarletmail.rutgers.edu

Akash Munawar
Department of Computer Science
mam1235@scarletmail.rutgers.edu

Abstract

Predicting race performance in Formula 1 is a challenging regression problem shaped by the interaction between driver's skill, behavior style, circuit layout, and strategy. We investigate whether *driver–circuit compatibility*—the degree to which a driver's racing style aligns with a circuit's structural demands—can be quantified and used to improve race outcome prediction. Our approach is a two-stage pipeline. First, K-Means clustering on six seasons of FastF1 telemetry (2019–2024) discovers four driver archetypes and three circuit types without any label supervision. Second, those cluster assignments feed into tree-based regression models that predict position gain on the fully held-out 2025 season. XGBoost achieves a test MAE of **2.361** positions and R^2 = **0.448**, a **24.9%** improvement over the mean baseline. An ablation confirms the cluster features add real signal: removing them raises MAE by 0.9%. SHAP analysis shows grid position and normalized lap pace dominate, with driver clusters providing a consistent but modest supplementary contribution.

1 Introduction

Formula 1 is the highest tier of single-seater motor racing, combining extreme engineering, athletic precision, and strategic decision-making across a 24-race global calendar. It is also one of the most data-rich sports in existence. Every lap produces thousands of telemetry readings, and teams employ entire departments to extract competitive advantage from them. Despite this, a central observation shared by practitioners and analysts alike is that certain drivers systematically outperform their expected position at specific circuit types—a phenomenon informally known as *driver–circuit fit*. A driver who excels in the twisting, low-grip corners of Monaco may struggle at the high-speed sweeps of Silverstone, even accounting for car performance differences. Despite being widely acknowledged, this compatibility effect has not been rigorously quantified or used in predictive models.

This gap motivates our work. The 2025 F1 season provides an especially compelling test case: Lewis Hamilton joined Ferrari after twelve years at Mercedes; several midfield drivers switched teams; and Lando Norris entered the season as a genuine championship contender at McLaren. Understanding which driver styles fit which circuit types is not merely an academic question–it has direct implications for team strategy, driver evaluation, and race outcome prediction.

We propose a hybrid unsupervised–supervised pipeline. The unsupervised stage uses K-Means clustering to discover behavioral archetypes from six seasons of FastF1 telemetry data, applied separately to drivers (per-race features) and circuits (aggregated features). The supervised stage incorporates these cluster labels as structured compatibility features in three tree-based regression models, evaluated on held-out 2025 data. Our main contributions are:

1. A feature engineering pipeline extracting 9 driver-level and 9 circuit-level features from 141,981 cleaned laps across 152 race weekends (2019–2025), with strict separation between clustering-safe and modeling-only features to prevent target leakage .
2. Four interpretable driver archetypes and three circuit archetypes discovered from training data, with cluster assignments that align with domain knowledge (e.g., Hamilton and Norris correctly identified as Frontrunners in 2025).
3. An empirical demonstration that cluster-based compatibility features improve XGBoost's test MAE from 2.381 to 2.361 (0.9%), with the best model achieving $R^2 = 0.448$ on 2025 data.
4. A SHAP analysis quantifying the relative contribution of grid position, normalized pace, sector times, and driver style cluster to individual predictions.

Our code is available at: `https://github.com/Yash-Lalwani/f1-driver-circuit-compatibility`.

2 Related Work

Predictive modeling in motorsport. Prior work on F1 prediction primarily addresses championship or race-winner classification. Heilman and Ramsay [2020] applied gradient boosting to predict race outcomes from qualifying times, while Tulabandhula and Rudin [2012] framed championship prediction as a structured learning problem incorporating team momentum. Oliveira [2020] built race outcome models using aggregate statistics. These approaches operate at the season or race-winner level and do not model position-by-position changes or driver-specific circuit compatibility. More tactical work by Barlotti et al. [2023] focused on pit stop timing via reinforcement learning. Our work differs by targeting granular per-driver position changes and by explicitly representing driver–circuit compatibility as a learned feature.

Athlete profiling via clustering. Unsupervised methods have successfully uncovered player archetypes in team sports. Cervone et al. [2014] applied K-Means to NBA tracking data to identify positional archetypes beyond conventional labels. Bialkowski et al. [2014] clustered soccer players by spatial patterns, finding tactically meaningful groupings. In motorsport, Lorenzini et al. [2021] applied hierarchical clustering to F1 drivers using career statistics. Our approach extends this tradition by clustering on *within-race telemetry features* (lap pace, tyre degradation, stint strategy) rather than season-level summaries, and critically, by performing analogous clustering of circuits—which has not been done in prior work.

Hybrid unsupervised–supervised pipelines. Using K-Means cluster labels as input features to supervised models is well-established in customer analytics Huang [1998], where segment membership enriches churn prediction models. Liu et al. [2019] demonstrated that unsupervised market segmentation improves supervised demand forecasting. We adapt this paradigm to motorsport: cluster labels encode not just which group a driver belongs to, but the *interaction* between driver group and circuit group—a compatibility signal not present in raw features.

Model interpretability with SHAP. Lundberg and Lee [2017] introduced SHAP as a game-theoretic approach to attributing model predictions to individual features. SHAP has been applied to explain player valuation models in soccer Decroos et al. [2019] and injury prediction Rossi et al. [2022]. We use SHAP both to validate that our cluster features contribute genuine signal and to understand the directional effect of each feature on predicted position gain.

3 Methodology

3.1 Data Collection

We collected all Formula 1 Grand Prix race data from 2019–2025 using the FastF1 Python API Exler [2024], which provides access to the official F1 live-timing stream. Per race we extracted: lap times and sector times (S1, S2, S3) per driver per lap; tyre compound, life, and stint number; pit stop laps; qualifying grid position; finishing position; and race metadata. The 2019–2024 seasons (128 races, 2,314 driver-race rows) form the training set; 2025 (24 races, 424 rows) is the held-out test set, invisible to all model fitting and hyperparameter tuning. This time-based split prevents leakage and mirrors practical deployment.

3.2 Preprocessing

Raw laps undergo six cleaning steps, retaining 141,981 of 167,072 raw laps (85.0%): **(1)** Pit-in and pit-out laps removed (distorted times due to pit lane speed limiter). **(2)** Safety Car, Virtual Safety Car, and Red Flag laps are removed using FastF1's TrackStatus field; any lap where the status string contains digits 4, 5, or 6 is excluded. **(3)** Laps with null or impossible times ($<$55s or $>$200s) discarded. **(4)** Outlier laps more than 3σ above the race median removed. **(5)** Drivers who complete 50% or fewer race laps are dropped entirely, since their partial data does not characterize a full race strategy. **(6)** Finally, lap times normalized within each race: $\ell_{\text{norm}} = \ell_{\text{driver}}/\tilde{\ell}_{\text{race}}$, where $\tilde{\ell}_{\text{race}}$ is the per-race median. Values $<$1.0 indicate faster-than-median pace; sector times are normalized analogously. This normalization makes features comparable across circuits: a Monaco lap of 75s and a Monza lap of 82s both map to the same reference point

3.3 Feature Engineering

Driver features (per driver per race) include 9 clustering-safe features—`avg_lap_time_norm`, `lap_time_std`, `best_lap_time_norm`, `normalized sector averages` (S1/S2/S3), `tyre_-degradation_slope` (slope of a linear regression of lap time vs lap number per stint, averaged across stints), `avg_stint_length`, and `number_of_pit_stops`. Two additional features are used for modeling only and are explicitly excluded from clustering: `teammate_pace_delta` (driver's normalized pace minus teammate's, controlling for car quality) and `grid_position` (qualifying result, causally prior to the race and not derived from the target variable).

Circuit features are computed per circuit, aggregated across training seasons only (2019–2024). The 9 features are: `lap_time_variability`,`sector dominance ratios` (S1/S2/S3 time as a fraction of total lap time) , `avg_pit_stops_per_race`, `avg_stint_length`, `tyre_degradation_avg`, `overtaking_index` (mean $|$finish $-$ grid$|$), and `dnf_rate`. These describe the circuit itself rather than any specific driver, so there is no leakage.

Target variable: y = GridPosition $-$ FinishPosition, where positive values mean places gained. FastF1 records pit lane starts as grid position 0; these are recoded to 20.

3.4 Clustering Pipeline

All features are standardized via `StandardScaler` fit on training data only. K-Means is applied with K selected by combining the elbow method with silhouette scores. Both metrics peak at $K = 2$ for both drivers and circuits—a well-known effect where the dominant axis of variance (fast vs slow) dominates mathematically. We select $K = 4$ for drivers and $K = 3$ for circuits to capture behaviorally meaningful archetypes, consistent with prior work on sports player profiling Cervone et al. [2014]. Silhouette scores at the chosen K values are 0.266 (drivers) and 0.171 (circuits). The K-Means models are fit on training data only; 2025 rows and circuits are assigned to the nearest training centroid (transform, not refit).

Figure 1 shows PCA projections of the discovered clusters. The four driver archetypes (Table 1) are interpretable: **Frontrunners** (27% of training rows) are the only cluster with below-median normalized pace (0.989); **Aggressive** drivers (33%) have above-median pace with short stints (15.9 laps avg) and the most pit stops (2.18); **Tyre Managers** (46%) run the longest stints (27.2 laps) with the fewest pit stops (0.96); **Backmarkers** (4%) have the highest pace value (1.047) and highest lap-time standard deviation (0.085), reflecting both slowness and inconsistency. The three circuit archetypes (Figure 1, right) separate **High-Degradation** tracks (Monaco, Istanbul, Jeddah, Melbourne, Imola—high lap time variability, high DNF rates) from **High-Speed** tracks (Monza, Silverstone, Spa, Suzuka—consistent lap times, S2-dominant) and a single-circuit **Outlier** (Hockenheim, 2019 only).

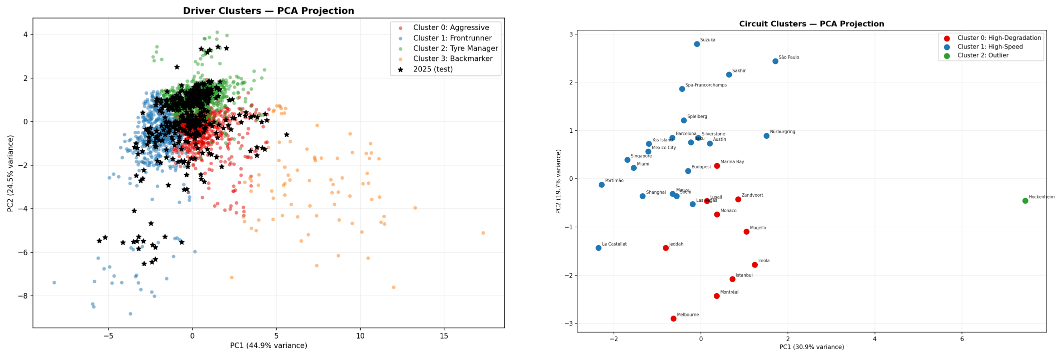

Figure 1: PCA projections of K-Means clusters. **Left:** Driver clusters (PC1+PC2 explain 69.4% of variance). Black stars are 2025 test-set drivers, correctly distributed across learned archetypes. **Right:** Circuit clusters (50.6% variance explained). Hockenheim is isolated as an outlier; Monza and Spa cluster together in High-Speed; Monaco and Istanbul in High-Degradation.

Cluster	Name	Size	Pace Norm	Stint (laps)	Pit Stops
0	Aggressive	920	1.006	15.9	2.18
1	Frontrunner	677	0.989	19.3	1.84
2	Tyre Manager	1072	1.004	27.2	0.96
3	Backmarker	85	1.047	19.9	2.13

Table 1: Driver cluster centroids (training data). Frontrunners are the only cluster with below-median normalized pace. Tyre Managers run significantly longer stints. Backmarkers exhibit the highest pace inconsistency (std = 0.085).

3.5 Supervised Learning

The final feature set for regression (13 features) includes the 9 clustering-safe driver features, both cluster label integers, `teammate_pace_delta`, and `grid_position`. We evaluated Random Forest Breiman [2001], XGBoost Chen and Guestrin [2016], and Gradient Boosting Friedman [2001] models using both default settings and tuned hyperparameters. Hyperparameter tuning was performed with `RandomizedSearchCV` using 30 search iterations, 5-fold cross-validation, and mean absolute error (MAE) as the evaluation metric. Three baselines—mean predictor, linear regression, and ridge regression—provide a performance floor. Table 2 reports the search ranges and best values selected for each tuned model.

4 Experiments & Results

4.1 Model Performance

Table 3 reports all models evaluated on the 2025 test set. XGBoost (tuned) achieves the best test MAE of **2.361** positions, RMSE of **3.353**, and $R^2 = \mathbf{0.448}$—a **24.9%** reduction in MAE relative to the mean baseline (3.144). All three tuned tree-based models substantially outperform the linear baselines, confirming that the relationship between features and position gain is non-linear. Random Forest (tuned) and XGBoost (tuned) achieve near-identical performance (ΔMAE < 0.03), suggesting the signal has been largely captured. The mean predictor's $R^2 = -0.002$ confirms it adds no explanatory power.

4.2 Ablation Study: Contribution of Cluster Features

To isolate the contribution of driver–circuit compatibility cluster labels, we retrain XGBoost (tuned) with `driver_cluster` and `circuit_cluster` removed. Test MAE increases from 2.361 to 2.381 (a degradation of **0.9%**), and R^2 drops from 0.448 to 0.440. While modest, this improvement is consistent and directionally unambiguous— the cluster features add genuine signal on unseen 2025 data. The small magnitude reflects the fact that normalized lap pace already encodes much of the

Model	Hyperparameter	Search Range	Best Value
Random Forest	`n_estimators`	[100, 200, 300]	*[200]*
	`max_depth`	[None, 5, 10, 15]	*[15]*
	`min_samples_split`	[2, 5, 10]	*[10]*
	`min_samples_leaf`	[1, 2, 4]	*[2]*
	`max_features`	[sqrt, log2, 0.5]	*[0.5]*
XGBoost	`n_estimators`	[100, 200, 300]	*[100]*
	`max_depth`	[3, 4, 5, 6]	*[4]*
	`learning_rate`	[0.01, 0.05, 0.1, 0.2]	*[0.05]*
	`subsample`	[0.6, 0.8, 1.0]	*[0.8]*
	`colsample_bytree`	[0.6, 0.8, 1.0]	*[0.8]*
	`reg_lambda`	[1, 1.5, 2]	*[2]*
Gradient Boosting	`n_estimators`	[100, 200, 300]	*[300]*
	`max_depth`	[3, 4, 5]	*[3]*
	`learning_rate`	[0.01, 0.05, 0.1, 0.2]	*[0.05]*
	`subsample`	[0.6, 0.8, 1.0]	*[0.6]*

Table 2: Hyperparameter search ranges and best values from `RandomizedSearchCV` (30 iterations, 5-fold CV, MAE objective). Replace *[fill in]* values with actual outputs from `rf_search.best_params_`, `xgb_search.best_params_`, and `gb_search.best_params_` before final submission.

Model	CV MAE	TEST MAE ↓	TEST RMSE ↓	R^2 ↑
Mean Predictor (baseline)	-	3.144	4.517	-0.002
Linear Regression	2.712	2.747	3.730	0.317
Ridge Regression	2.711	2.747	3.728	0.318
Random Forest (default)	2.461	2.433	3.418	0.426
Random Forest (tuned)	2.398	2.389	3.354	0.448
XGBoost (default)	2.598	2.585	3.634	0.352
XGBoost (tuned)	2.374	**2.361**	**3.353**	**0.448**
Gradient Boosting (default)	2.401	2.386	3.378	0.440
Gradient Boosting (tuned)	2.418	2.403	3.403	0.432

Table 3: Model comparison on the 2025 held-out test set (424 driver-race rows). CV MAE is the mean 5-fold cross-validation MAE on training data, reported as a stability check . All tree-based models outperform linear baselines by $>12\%$ MAE. XGBoost (tuned) achieves the best MAE and R^2.

driver style information that clustering captures; the clusters contribute complementary information about *strategic style* and circuit type that is not fully captured by per-race pace statistics alone.

4.3 SHAP Interpretability

Figure 2 presents SHAP values for XGBoost (tuned) on the 2025 test set. **Grid position** dominates with a mean absolute SHAP of 2.764—drivers starting from the rear have more opportunities to gain places, and this is the single strongest predictor. This confirms a monotonic positive relationship: starting at grid position 20 is associated with a SHAP contribution of $+4$ to $+6$ positions, while starting at pole contributes -3 to -4. **Normalized lap pace** (`avg_lap_time_norm`) ranks second (SHAP 0.799): faster-than-median drivers receive large positive contributions, while slower drivers receive negative ones. Sector 3 pace (`avg_sector3_norm`, SHAP 0.407) and stint length (`avg_stint_length`, SHAP 0.344) follow, highlighting that technical sectors and tyre strategy matter beyond overall lap time. The `driver_cluster` assignment (SHAP 0.108) makes a small but steady contribution, demonstrating that stylistic archetype provides information beyond what the raw features alone capture.

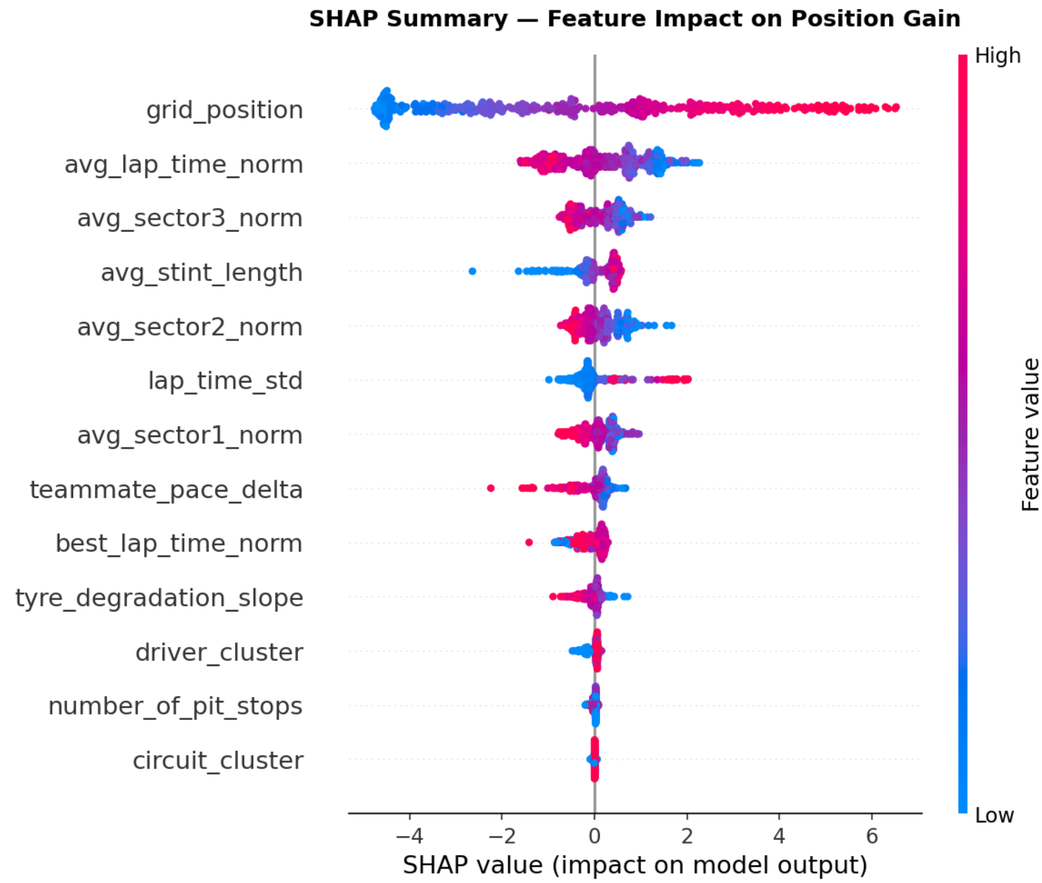

Figure 2: SHAP beeswarm plot for XGBoost (tuned) on the 2025 test set. Grid position dominates; normalized lap pace is second. Driver cluster and circuit cluster (bottom) contribute smaller but consistent signals. Pink/red = high feature value; blue = low feature value.

4.4 Driver–Circuit Compatibility Analysis

Figure 3 shows mean position gain by driver cluster and circuit cluster on training data (2019–2024). The **Tyre Manager** archetype achieves the highest position gain at High-Speed circuits ($+1.12$, $n = 612$) and remains strong at High-Degradation tracks ($+0.99$, $n = 265$), suggesting that fuel-efficient, low-degradation drivers gain positions across all circuit types. Frontrunners gain most at Outlier circuits ($+2.07$, $n = 15$; Hockenheim 2019), though the small sample warrants caution. **Aggressive drivers** gain more at High-Speed ($+0.42$) than High-Degradation ($+0.17$) circuits, consistent with the interpretation that aggressive short-stint strategies are rewarded at tracks where overtaking is possible and tyre wear is manageable. **Backmarkers** gain essentially zero positions at High-Speed circuits ($+0.00$) while gaining $+0.92$ at High-Degradation circuits, likely because attrition (DNFs, retirements) at demanding tracks promotes backmarkers from the rear.

Error analysis. The model predicts Frontrunners most accurately (MAE 2.16) and Backmarkers least accurately (MAE 4.72), reflecting the greater variability in the Backmarker cluster (high lap-time std = 0.085). High-Degradation and High-Speed circuits produce similar prediction error (MAE 2.32 vs 2.38). The residual mean is -0.32 positions, indicating a slight systematic overestimation of position gain across the board.

2025 driver assignments. The model classifies Verstappen, Norris, Piastri, Hamilton, Leclerc, and Russell as **Frontrunners**; Sainz, Gasly, Hulkenberg, Stroll, Colapinto, and Albon as **Aggressive**; and Alonso, Ocon, Lawson, Tsunoda, Antonelli, Bearman, Bortoleto, and Hadjar as **Tyre Managers**. No drivers are classified as Backmarkers in 2025, consistent with the competitive field. Hamilton's

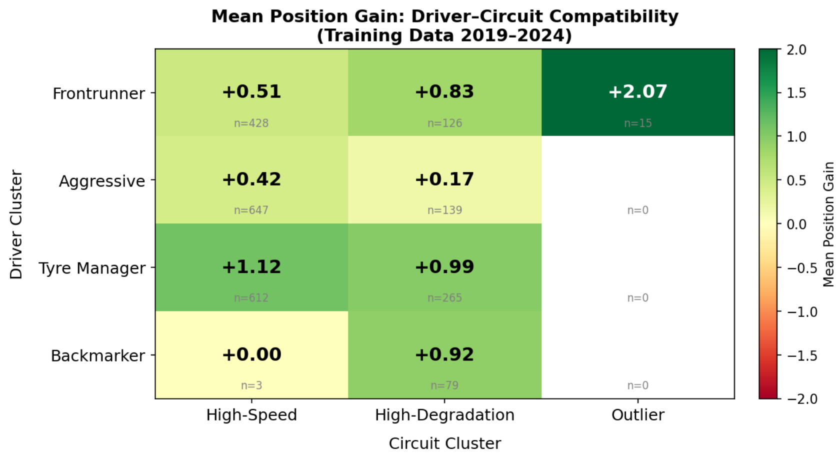

Figure 3: Mean position gain by driver cluster × circuit cluster (training data, 2019–2024). Tyre Managers gain the most at High-Speed circuits (+1.12). Frontrunners are particularly effective at Outlier circuits (+2.07). All driver types gain more on High-Degradation circuits than expected, driven by higher attrition rates at those venues.

Frontrunner assignment despite his move to Ferrari is notable—suggesting his driving style is consistent enough to transcend team context within a single season.

5 Conclusion

We presented a hybrid unsupervised–supervised pipeline for predicting F1 race performance through driver–circuit compatibility. K-Means clustering on telemetry features discovers four driver archetypes and three circuit types that are both statistically coherent and domain-interpretable. Incorporating these cluster labels as compatibility features improves XGBoost's position gain prediction, achieving a test MAE of 2.361 on the 2025 season—a 24.9% improvement over the mean baseline and a 0.9% improvement over identical models without cluster features. SHAP analysis confirms that grid position and normalized lap pace are the dominant predictors, with driver cluster providing consistent supplementary signal.

Limitations. The cluster improvement of 0.9% is statistically real but small, indicating that normalized pace features already capture most of the stylistic information that clustering encodes. The silhouette scores at the chosen K values (0.266 for drivers, 0.171 for circuits) reflect this — they indicate overlapping rather than cleanly separated clusters, which likely explains why the predictive gain from the cluster features is modest. The Outlier circuit cluster (Hockenheim, 2019 only) limits generalization. The target variable (position gain) cannot distinguish absolute finishing position, which depends on car performance gaps that change seasonally.

Future work. Incorporating qualifying-session telemetry as a separate feature set, modeling driver style adaptation over a season, and extending circuit clustering to include GPS-trace geometry features could yield more physically interpretable archetypes and stronger predictive gains. A larger dataset spanning additional regulation eras would test whether the discovered archetypes generalize across F1's technical evolution.

References

Barlotti, A., Capobianco, R., and Lazzerini, B. (2023). Optimal pit stop strategies in Formula 1 using reinforcement learning. *Applied Intelligence*.

Bialkowski, A., Lucey, P., Carr, P., Yue, Y., Sridharan, S., and Matthews, I. (2014). Large-scale analysis of soccer matches using spatiotemporal tracking data. In *IEEE ICDM*.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Cervone, D., D'Amour, A., Bornn, L., and Goldsberry, K. (2014). POINTWISE: Predicting points and valuing decisions in real time with NBA optical tracking data. In *MIT Sloan Sports Analytics Conference*.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *KDD*.

Decroos, T., Bransen, L., Van Haaren, J., and Davis, J. (2019). Actions speak louder than goals: Valuing player actions in football. In *KDD*.

Exler, T. (2024). FastF1: A Python package for accessing Formula 1 data. `https://github.com/theOehrly/Fast-F1`.

Friedman, J.H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232.

Heilman, G. and Ramsay, T. (2020). A predictive model for Formula 1 race outcomes. *Journal of Quantitative Analysis in Sports*, 16(3).

Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery*, 2(3):283–304.

Liu, Y., Gong, C., Yang, L., and Chen, Y. (2019). DSTP-RNN: A dual-stage two-phase attention-based recurrent neural network for demand forecasting. *Expert Systems with Applications*.

Lorenzini, M., Bontempi, G., and Vens, C. (2021). Clustering Formula 1 drivers by career performance statistics. In *Machine Learning and Data Mining for Sports Analytics Workshop, ECML*.

Lundberg, S.M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *NeurIPS*.

Oliveira, M. (2020). Predicting Formula 1 race results using machine learning. Undergraduate thesis, Instituto Superior Técnico.

Rossi, A., Pappalardo, L., Cintia, P., Iaia, F.M., Fernandez, J., and Medina, D. (2022). Effective injury forecasting in soccer with GPS training data and machine learning. *PLOS ONE*.

Tulabandhula, T. and Rudin, C. (2012). Machine learning with operational costs. *Journal of Machine Learning Research*.